

`./test.sh 10 5 1 0 25:`

```
~/cse330-files-project2 # ./test.sh 10 5 1 0 25
Checking passed arguments
/dev/null exists; moving on..
rm process_generator
gcc -o process_generator process_generator.c
Test script will start 10 processes for the user 1000
UID:1000 Process-1 Pid 303
UID:1000 Process-2 Pid 304
UID:1000 Process-3 Pid 305
UID:1000 Process-4 Pid 306
UID:1000 Process-5 Pid 307
UID:1000 Process-6 Pid 308
UID:1000 Process-7 Pid 309
UID:1000 Process-8 Pid 310
UID:1000 Process-9 Pid 311
Continuing ..
Test script will insert the test kernel module
[ 67.842026] producer_consumer: loading out-of-tree module taints kernel.
[ 67.845682] ###[thread_init_module]###CSE330 Project Kernel Module Inserted
[ 67.847463] ###[thread_init_module]###Kernel module received these inputs: UID:1000, Buffer-Si0
[ 67.850358] ###[producer_thread_function]###[kProducer-1] Produce-Item#:1 at buffer index: 4 f
[ 67.852514] ###[producer_thread_function]###[kProducer-1] Produce-Item#:2 at buffer index: 4 f
[ 67.854893] ###[producer_thread_function]###[kProducer-1] Produce-Item#:3 at buffer index: 4 f
The test script waits 10 seconds for the kernel module to finish
[ 67.857074] ###[producer_thread_function]###[kProducer-1] Produce-Item#:4 at buffer index: 4 f
[ 67.860766] ###[producer_thread_function]###[kProducer-1] Produce-Item#:5 at buffer index: 4 f
[ 67.862931] ###[producer_thread_function]###[kProducer-1] Produce-Item#:6 at buffer index: 4 f
[ 67.865230] ###[producer_thread_function]###[kProducer-1] Produce-Item#:7 at buffer index: 4 f
[ 67.867440] ###[producer_thread_function]###[kProducer-1] Produce-Item#:8 at buffer index: 4 f
[ 67.869688] ###[producer_thread_function]###[kProducer-1] Produce-Item#:9 at buffer index: 4 f
[ 67.871895] ###[producer_thread_function]###[kProducer-1] Produce-Item#:10 at buffer index: 4
[ 67.874104] ###[producer_thread_function]###[kProducer-1] Producer Thread stopped.
Test script will remove the test kernel module
```

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[ 67.867440] ###[producer_thread_function]###[kProducer-1] Produce-Item#:8 at buffer index: 4 f
[ 67.869688] ###[producer_thread_function]###[kProducer-1] Produce-Item#:9 at buffer index: 4 f
[ 67.871895] ###[producer_thread_function]###[kProducer-1] Produce-Item#:10 at buffer index: 4
[ 67.874104] ###[producer_thread_function]###[kProducer-1] Producer Thread stopped.
Test script will remove the test kernel module
[ 77.858889] ###[thread_exit_module]###Total number of items produced: 10
[ 77.858899] ###[thread_exit_module]###Total number of items consumed: 0
[ 77.860577] ###[thread_exit_module]###The total elapsed time UID 1000 is 0:0:0
[ 77.864146] ###[thread_exit_module]###CSE330 Project Kernel Module Removed
The following is the dmesg output
[ 3.015509] mount (67) used greatest stack depth: 13904 bytes left
[ 3.352303] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/i8042/serio1/input/inp3
[ 6.212507] make (68) used greatest stack depth: 13576 bytes left
[ 25.430353] cc1 (88) used greatest stack depth: 13520 bytes left
[ 25.747587] as (95) used greatest stack depth: 13280 bytes left
[ 26.533680] random: crng init done
[ 35.669716] sed (276) used greatest stack depth: 13120 bytes left
[ 67.842026] producer_consumer: loading out-of-tree module taints kernel.
[ 67.845682] ###[thread_init_module]###CSE330 Project Kernel Module Inserted
[ 67.847463] ###[thread_init_module]###Kernel module received these inputs: UID:1000, Buffer-Si0
[ 67.850358] ###[producer_thread_function]###[kProducer-1] Produce-Item#:1 at buffer index: 4 f
[ 67.852514] ###[producer_thread_function]###[kProducer-1] Produce-Item#:2 at buffer index: 4 f
[ 67.854893] ###[producer_thread_function]###[kProducer-1] Produce-Item#:3 at buffer index: 4 f
[ 67.857074] ###[producer_thread_function]###[kProducer-1] Produce-Item#:4 at buffer index: 4 f
[ 67.860766] ###[producer_thread_function]###[kProducer-1] Produce-Item#:5 at buffer index: 4 f
[ 67.862931] ###[producer_thread_function]###[kProducer-1] Produce-Item#:6 at buffer index: 4 f
[ 67.865230] ###[producer_thread_function]###[kProducer-1] Produce-Item#:7 at buffer index: 4 f
[ 67.867440] ###[producer_thread_function]###[kProducer-1] Produce-Item#:8 at buffer index: 4 f
[ 67.869688] ###[producer_thread_function]###[kProducer-1] Produce-Item#:9 at buffer index: 4 f
[ 67.871895] ###[producer_thread_function]###[kProducer-1] Produce-Item#:10 at buffer index: 4
[ 67.874104] ###[producer_thread_function]###[kProducer-1] Producer Thread stopped.
[ 77.858889] ###[thread_exit_module]###Total number of items produced: 10
[ 77.858899] ###[thread_exit_module]###Total number of items consumed: 0
[ 77.860577] ###[thread_exit_module]###The total elapsed time UID 1000 is 0:0:0
[ 77.864146] ###[thread_exit_module]###CSE330 Project Kernel Module Removed
Total Elapsed Time from PS Command 0:1:4
~/cse330-files-project2 #

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./test.sh 10 5 0 1 25:

```
~/cse330-files-project2 # ./test.sh 10 5 0 1 25
Checking passed arguments
useradd: user 'test_cse330' already exists
/dev/null exists; moving on..
rm process_generator
gcc -o process_generator process_generator.c
Test script will start 10 processes for the user 1000
UID:1000 Process-1 Pid 1857
UID:1000 Process-2 Pid 1858
UID:1000 Process-3 Pid 1859
UID:1000 Process-4 Pid 1860
UID:1000 Process-5 Pid 1861
UID:1000 Process-6 Pid 1862
UID:1000 Process-7 Pid 1863
UID:1000 Process-8 Pid 1864
UID:1000 Process-9 Pid 1865
Continuing ..
Test script will insert the test kernel module
[ 982.957880] ###[thread_init_module]###CSE330 Project Kernel Module Inserted
[ 982.959429] ###[thread_init_module]###Kernel module received these inputs: UID:1000, Buffer-Si1
The test script waits 10 seconds for the kernel module to finish
Test script will remove the test kernel module
[ 992.965347] ###[consumer_thread_function]###[kConsumer-1] Consumer Thread stopped.
[ 992.967460] ###[thread_exit_module]###Total number of items produced: 0
[ 992.967466] ###[thread_exit_module]###Total number of items consumed: 0
[ 992.969403] ###[thread_exit_module]###The total elapsed time UID 1000 is 0:0:0
[ 992.973501] ###[thread_exit_module]###CSE330 Project Kernel Module Removed
The following is the dmesg output
[ 878.010537] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-488 on buffer index:3
[ 878.013620] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-489 on buffer index:3
[ 878.016632] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-490 on buffer index:3
[ 878.019625] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-491 on buffer index:4
[ 878.022674] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-492 on buffer index:4
[ 878.025838] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-493 on buffer index:4
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[ 982.957880] ####[thread_init_module]###CSE330 Project Kernel Module Inserted
[ 982.959429] ####[thread_init_module]###Kernel module received these inputs: UID:1000, Buffer-Si1
The test script waits 10 seconds for the kernel module to finish
Test script will remove the test kernel module
[ 992.965347] ####[consumer_thread_function]###[kConsumer-1] Consumer Thread stopped.
[ 992.967460] ####[thread_exit_module]###Total number of items produced: 0
[ 992.967466] ####[thread_exit_module]###Total number of items consumed: 0
[ 992.969403] ####[thread_exit_module]###The total elapsed time UID 1000 is 0:0:0
[ 992.973501] ####[thread_exit_module]###CSE330 Project Kernel Module Removed
The following is the dmesg output
[ 878.010537] ####[consumer_thread_function]###[kConsumer-1] Consumed Item#-488 on buffer index:3
[ 878.013620] ####[consumer_thread_function]###[kConsumer-1] Consumed Item#-489 on buffer index:3
[ 878.016632] ####[consumer_thread_function]###[kConsumer-1] Consumed Item#-490 on buffer index:3
[ 878.019625] ####[consumer_thread_function]###[kConsumer-1] Consumed Item#-491 on buffer index:4
[ 878.022674] ####[consumer_thread_function]###[kConsumer-1] Consumed Item#-492 on buffer index:4
[ 878.025838] ####[consumer_thread_function]###[kConsumer-1] Consumed Item#-493 on buffer index:4
[ 878.028872] ####[consumer_thread_function]###[kConsumer-1] Consumed Item#-494 on buffer index:4
[ 878.031862] ####[consumer_thread_function]###[kConsumer-1] Consumed Item#-495 on buffer index:4
[ 878.034879] ####[consumer_thread_function]###[kConsumer-1] Consumed Item#-496 on buffer index:4
[ 878.037884] ####[consumer_thread_function]###[kConsumer-1] Consumed Item#-497 on buffer index:4
[ 878.040923] ####[consumer_thread_function]###[kConsumer-1] Consumed Item#-498 on buffer index:4
[ 878.043929] ####[consumer_thread_function]###[kConsumer-1] Consumed Item#-499 on buffer index:4
[ 878.047039] ####[consumer_thread_function]###[kConsumer-1] Consumed Item#-500 on buffer index:4
[ 885.440527] ####[consumer_thread_function]###[kConsumer-1] Consumer Thread stopped.
[ 885.442325] ####[thread_exit_module]###Total number of items produced: 500
[ 885.442330] ####[thread_exit_module]###Total number of items consumed: 500
[ 885.444016] ####[thread_exit_module]###The total elapsed time UID 1000 is 1:17:47
[ 885.447481] ####[thread_exit_module]###CSE330 Project Kernel Module Removed
[ 982.957880] ####[thread_init_module]###CSE330 Project Kernel Module Inserted
[ 982.959429] ####[thread_init_module]###Kernel module received these inputs: UID:1000, Buffer-Si1
[ 992.965347] ####[consumer_thread_function]###[kConsumer-1] Consumer Thread stopped.
[ 992.967460] ####[thread_exit_module]###Total number of items produced: 0
[ 992.967466] ####[thread_exit_module]###Total number of items consumed: 0
[ 992.969403] ####[thread_exit_module]###The total elapsed time UID 1000 is 0:0:0
[ 992.973501] ####[thread_exit_module]###CSE330 Project Kernel Module Removed
Total Elapsed Time from PS Command 0:1:4
~/cse330-files-project2 #

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`./test.sh 10 50 1 1 25:`

```
~/cse330-files-project2 # ./test.sh 10 50 1 1 25
Checking passed arguments
useradd: user 'test_cse330' already exists
/dev/null exists; moving on..
rm process_generator
gcc -o process_generator process_generator.c
Test script will start 10 processes for the user 1000
UID:1000 Process-1 Pid 377
UID:1000 Process-2 Pid 378
UID:1000 Process-3 Pid 379
UID:1000 Process-4 Pid 380
UID:1000 Process-5 Pid 381
UID:1000 Process-6 Pid 382
UID:1000 Process-7 Pid 383
UID:1000 Process-8 Pid 384
UID:1000 Process-9 Pid 385
Continuing ..
[ 224.264245] ps (389) used greatest stack depth: 12912 bytes left
Test script will insert the test kernel module
[ 224.272531] ###[thread_init_module]###CSE330 Project Kernel Module Inserted
[ 224.274121] ###[thread_init_module]###Kernel module received these inputs: UID:1000, Buffer-Si1
[ 224.276752] ###[producer_thread_function]###[kProducer-1] Produce-Item#:1 at buffer index: 0 f
[ 224.276771] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-1 on buffer index:0::
[ 224.278746] ###[producer_thread_function]###[kProducer-1] Produce-Item#:2 at buffer index: 1 f
[ 224.283725] ###[producer_thread_function]###[kProducer-1] Produce-Item#:3 at buffer index: 2 f
[ 224.283755] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-2 on buffer index:1::
[ 224.285830] ###[producer_thread_function]###[kProducer-1] Produce-Item#:4 at buffer index: 3 f
[ 224.288521] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-3 on buffer index:2::
[ 224.290986] ###[producer_thread_function]###[kProducer-1] Produce-Item#:5 at buffer index: 4 f
[ 224.295693] ###[producer_thread_function]###[kProducer-1] Produce-Item#:6 at buffer index: 5 f
[ 224.297672] ###[producer_thread_function]###[kProducer-1] Produce-Item#:7 at buffer index: 6 f
[ 224.299563] ###[producer_thread_function]###[kProducer-1] Produce-Item#:8 at buffer index: 7 f
The test script [ 224.301415] ###[producer_thread_function]###[kProducer-1] Produce-Item#:9 at b
waits 10 seconds[ 224.303568] ###[producer_thread_function]###[kProducer-1] Produce-Item#:10 at
for the kernel [ 224.305730] ###[producer_thread_function]###[kProducer-1] Producer Thread stop.
```



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[ 224.295693] ###[producer_thread_function]###[kProducer-1] Produce-Item#:6 at buffer index: 5 f
[ 224.297672] ###[producer_thread_function]###[kProducer-1] Produce-Item#:7 at buffer index: 6 f
[ 224.299563] ###[producer_thread_function]###[kProducer-1] Produce-Item#:8 at buffer index: 7 f
The test script [ 224.301415] ###[producer_thread_function]###[kProducer-1] Produce-Item#:9 at b
waits 10 seconds[ 224.303568] ###[producer_thread_function]###[kProducer-1] Produce-Item#:10 at
for the kernel [ 224.305730] ###[producer_thread_function]###[kProducer-1] Producer Thread stop.
module to finish[ 224.308295] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-4 on
[ 224.311273] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-5 on buffer index:4::

[ 224.313676] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-6 on buffer index:5::
[ 224.316435] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-7 on buffer index:6::
[ 224.318772] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-8 on buffer index:7::
[ 224.321225] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-9 on buffer index:8::
[ 224.323667] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-10 on buffer index:9:
Test script will remove the test kernel module
[ 234.321321] ###[consumer_thread_function]###[kConsumer-1] Consumer Thread stopped.
[ 234.323424] ###[thread_exit_module]###Total number of items produced: 10
[ 234.323429] ###[thread_exit_module]###Total number of items consumed: 10
[ 234.325218] ###[thread_exit_module]###The total elapsed time UID 1000 is 0:1:5
[ 234.328905] ###[thread_exit_module]###CSE330 Project Kernel Module Removed
The following is the dmesg output
[ 224.276771] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-1 on buffer index:0::
[ 224.278746] ###[producer_thread_function]###[kProducer-1] Produce-Item#:2 at buffer index: 1 f
[ 224.283725] ###[producer_thread_function]###[kProducer-1] Produce-Item#:3 at buffer index: 2 f
[ 224.283755] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-2 on buffer index:1::
[ 224.285830] ###[producer_thread_function]###[kProducer-1] Produce-Item#:4 at buffer index: 3 f
[ 224.288521] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-3 on buffer index:2::
[ 224.290986] ###[producer_thread_function]###[kProducer-1] Produce-Item#:5 at buffer index: 4 f
[ 224.295693] ###[producer_thread_function]###[kProducer-1] Produce-Item#:6 at buffer index: 5 f
[ 224.297672] ###[producer_thread_function]###[kProducer-1] Produce-Item#:7 at buffer index: 6 f
[ 224.299563] ###[producer_thread_function]###[kProducer-1] Produce-Item#:8 at buffer index: 7 f
[ 224.301415] ###[producer_thread_function]###[kProducer-1] Produce-Item#:9 at buffer index: 8 f
[ 224.303568] ###[producer_thread_function]###[kProducer-1] Produce-Item#:10 at buffer index: 9
[ 224.305730] ###[producer_thread_function]###[kProducer-1] Producer Thread stopped.
[ 224.308295] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-4 on buffer index:3::
[ 224.311273] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-5 on buffer index:4::
[ 224.313676] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-6 on buffer index:5::

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[ 224.318772] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-8 on buffer index:7::
[ 224.321225] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-9 on buffer index:8::
[ 224.323667] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-10 on buffer index:9:
Test script will remove the test kernel module
[ 234.321321] ###[consumer_thread_function]###[kConsumer-1] Consumer Thread stopped.
[ 234.323424] ###[thread_exit_module]###Total number of items produced: 10
[ 234.323429] ###[thread_exit_module]###Total number of items consumed: 10
[ 234.325218] ###[thread_exit_module]###The total elapsed time UID 1000 is      0:1:5
[ 234.328905] ###[thread_exit_module]###CSE330 Project Kernel Module Removed
The following is the dmesg output
[ 224.276771] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-1 on buffer index:0::
[ 224.278746] ###[producer_thread_function]###[kProducer-1] Produce-Item#:2 at buffer index: 1 f
[ 224.283725] ###[producer_thread_function]###[kProducer-1] Produce-Item#:3 at buffer index: 2 f
[ 224.283755] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-2 on buffer index:1::
[ 224.285830] ###[producer_thread_function]###[kProducer-1] Produce-Item#:4 at buffer index: 3 f
[ 224.288521] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-3 on buffer index:2::
[ 224.290986] ###[producer_thread_function]###[kProducer-1] Produce-Item#:5 at buffer index: 4 f
[ 224.295693] ###[producer_thread_function]###[kProducer-1] Produce-Item#:6 at buffer index: 5 f
[ 224.297672] ###[producer_thread_function]###[kProducer-1] Produce-Item#:7 at buffer index: 6 f
[ 224.299563] ###[producer_thread_function]###[kProducer-1] Produce-Item#:8 at buffer index: 7 f
[ 224.301415] ###[producer_thread_function]###[kProducer-1] Produce-Item#:9 at buffer index: 8 f
[ 224.303568] ###[producer_thread_function]###[kProducer-1] Produce-Item#:10 at buffer index: 9
[ 224.305730] ###[producer_thread_function]###[kProducer-1] Producer Thread stopped.
[ 224.308295] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-4 on buffer index:3::
[ 224.311273] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-5 on buffer index:4::
[ 224.313676] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-6 on buffer index:5::
[ 224.316435] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-7 on buffer index:6::
[ 224.318772] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-8 on buffer index:7::
[ 224.321225] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-9 on buffer index:8::
[ 224.323667] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-10 on buffer index:9:
[ 234.321321] ###[consumer_thread_function]###[kConsumer-1] Consumer Thread stopped.
[ 234.323424] ###[thread_exit_module]###Total number of items produced: 10
[ 234.323429] ###[thread_exit_module]###Total number of items consumed: 10
[ 234.325218] ###[thread_exit_module]###The total elapsed time UID 1000 is      0:1:5
[ 234.328905] ###[thread_exit_module]###CSE330 Project Kernel Module Removed
Total Elapsed Time from PS Command 0:1:4
~/cse330-files-project2 #

```

[./test.sh](#) 100 50 1 1 25:

```
~/cse330-files-project2 # ./test.sh 100 50 1 1 25
Checking passed arguments
useradd: user 'test_cse330' already exists
/dev/null exists; moving on..
rm process_generator
gcc -o process_generator process_generator.c
Test script will start 100 processes for the user 1000
UID:1000 Process-10 Pid 424
UID:1000 Process-5 Pid 419
UID:1000 Process-3 Pid 417
UID:1000 Process-1 Pid 415
UID:1000 Process-2 Pid 416
UID:1000 Process-8 Pid 422
UID:1000 Process-4 Pid 418
UID:1000 Process-6 Pid 420
UID:1000 Process-7 Pid 421
UID:1000 Process-9 Pid 423
UID:1000 Process-11 Pid 425
UID:1000 Process-12 Pid 426
UID:1000 Process-13 Pid 427
UID:1000 Process-14 Pid 428
UID:1000 Process-15 Pid 429
UID:1000 Process-16 Pid 430
UID:1000 Process-17 Pid 431
UID:1000 Process-18 Pid 432
UID:1000 Process-19 Pid 433
UID:1000 Process-20 Pid 434
UID:1000 Process-21 Pid 435
UID:1000 Process-22 Pid 436
UID:1000 Process-23 Pid 437
UID:1000 Process-24 Pid 438
UID:1000 Process-25 Pid 439
UID:1000 Process-26 Pid 440
UID:1000 Process-27 Pid 441
```

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UID:1000 Process-99 Pid 513
UID:1000 Process-95 Pid 509
UID:1000 Process-96 Pid 510
UID:1000 Process-98 Pid 512
Continuing ..
Test script will insert the test kernel module
[ 322.916456] ###[thread_init_module]###CSE330 Project Kernel Module Inserted
[ 322.918054] ###[thread_init_module]###Kernel module received these inputs: UID:1000, Buffer-Si1
[ 322.920993] ###[producer_thread_function]###[kProducer-1] Produce-Item#:1 at buffer index: 0 f
[ 322.923059] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-1 on buffer index:0::
[ 322.923082] ###[producer_thread_function]###[kProducer-1] Produce-Item#:2 at buffer index: 1 f
[ 322.925914] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-2 on buffer index:1::
[ 322.928018] ###[producer_thread_function]###[kProducer-1] Produce-Item#:3 at buffer index: 2 f
[ 322.930868] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-3 on buffer index:2::
[ 322.935311] ###[producer_thread_function]###[kProducer-1] Produce-Item#:4 at buffer index: 3 f
The test script [ 322.937434] ###[producer_thread_function]###[kProducer-1] Produce-Item#:5 at b
waits 10 seconds[ 322.939825] ###[producer_thread_function]###[kProducer-1] Produce-Item#:6 at b
for the kernel [ 322.942350] ###[producer_thread_function]###[kProducer-1] Produce-Item#:7 at b
module to finish[ 322.944699] ###[producer_thread_function]###[kProducer-1] Produce-Item#:8 at b
[ 322.945188] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-4 on buffer index:3::
[ 322.950112] ###[producer_thread_function]###[kProducer-1] Produce-Item#:9 at buffer index: 8 f
[ 322.950117] ###[producer_thread_function]###[kProducer-1] Produce-Item#:10 at buffer index: 9
[ 322.950119] ###[producer_thread_function]###[kProducer-1] Produce-Item#:11 at buffer index: 10
[ 322.950121] ###[producer_thread_function]###[kProducer-1] Produce-Item#:12 at buffer index: 11
[ 322.950123] ###[producer_thread_function]###[kProducer-1] Produce-Item#:13 at buffer index: 12
[ 322.950125] ###[producer_thread_function]###[kProducer-1] Produce-Item#:14 at buffer index: 13
[ 322.950127] ###[producer_thread_function]###[kProducer-1] Produce-Item#:15 at buffer index: 14

[ 322.952516] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-5 on buffer index:4::
[ 322.957309] ###[producer_thread_function]###[kProducer-1] Produce-Item#:16 at buffer index: 15
[ 322.966254] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-6 on buffer index:5::
[ 322.968485] ###[producer_thread_function]###[kProducer-1] Produce-Item#:17 at buffer index: 16
[ 322.970701] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-7 on buffer index:6::
[ 322.973021] ###[producer_thread_function]###[kProducer-1] Produce-Item#:18 at buffer index: 17
[ 322.975313] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-8 on buffer index:7::
[ 322.977578] ###[producer_thread_function]###[kProducer-1] Produce-Item#:19 at buffer index: 18
[ 322.979883] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-9 on buffer index:8::
```

```
[ 323.236293] ###[producer_thread_function]###[kProducer-1] Producer Thread stopped.
[ 323.238381] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-82 on buffer index:31
[ 323.355886] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-83 on buffer index:32
[ 323.358481] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-84 on buffer index:33
[ 323.361138] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-85 on buffer index:34
[ 323.363617] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-86 on buffer index:35
[ 323.366263] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-87 on buffer index:36
[ 323.368809] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-88 on buffer index:37
[ 323.371345] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-89 on buffer index:38
[ 323.373905] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-90 on buffer index:39
[ 323.376442] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-91 on buffer index:40
[ 323.378905] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-92 on buffer index:41
[ 323.381407] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-93 on buffer index:42
[ 323.383777] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-94 on buffer index:43
[ 323.386320] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-95 on buffer index:44
[ 323.388747] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-96 on buffer index:45
[ 323.391238] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-97 on buffer index:46
[ 323.393713] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-98 on buffer index:47
[ 323.396274] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-99 on buffer index:48
[ 323.398730] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-100 on buffer index:4
```

Test script will remove the test kernel module

```
[ 332.968574] ###[consumer_thread_function]###[kConsumer-1] Consumer Thread stopped.
[ 332.970936] ###[thread_exit_module]###Total number of items produced: 100
[ 332.970942] ###[thread_exit_module]###Total number of items consumed: 100
[ 332.972970] ###[thread_exit_module]###The total elapsed time UID 1000 is 0:9:45
[ 332.976427] ###[thread_exit_module]###CSE330 Project Kernel Module Removed
```

The following is the dmesg output

```
[ 323.236293] ###[producer_thread_function]###[kProducer-1] Producer Thread stopped.
[ 323.238381] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-82 on buffer index:31
[ 323.355886] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-83 on buffer index:32
[ 323.358481] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-84 on buffer index:33
[ 323.361138] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-85 on buffer index:34
[ 323.363617] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-86 on buffer index:35
[ 323.366263] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-87 on buffer index:36
[ 323.368809] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-88 on buffer index:37
[ 323.371345] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-89 on buffer index:38
[ 323.373905] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-90 on buffer index:39
```

```

[ 323.393713] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-98 on buffer index:47
[ 323.396274] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-99 on buffer index:48
[ 323.398730] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-100 on buffer index:4
Test script will remove the test kernel module
[ 332.968574] ###[consumer_thread_function]###[kConsumer-1] Consumer Thread stopped.
[ 332.970936] ###[thread_exit_module]###Total number of items produced: 100
[ 332.970942] ###[thread_exit_module]###Total number of items consumed: 100
[ 332.972970] ###[thread_exit_module]###The total elapsed time UID 1000 is 0:9:45
[ 332.976427] ###[thread_exit_module]###CSE330 Project Kernel Module Removed
The following is the dmesg output
[ 323.236293] ###[producer_thread_function]###[kProducer-1] Producer Thread stopped.
[ 323.238381] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-82 on buffer index:31
[ 323.355886] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-83 on buffer index:32
[ 323.358481] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-84 on buffer index:33
[ 323.361138] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-85 on buffer index:34
[ 323.363617] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-86 on buffer index:35
[ 323.366263] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-87 on buffer index:36
[ 323.368809] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-88 on buffer index:37
[ 323.371345] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-89 on buffer index:38
[ 323.373905] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-90 on buffer index:39
[ 323.376442] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-91 on buffer index:40
[ 323.378905] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-92 on buffer index:41
[ 323.381407] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-93 on buffer index:42
[ 323.383777] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-94 on buffer index:43
[ 323.386320] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-95 on buffer index:44
[ 323.388747] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-96 on buffer index:45
[ 323.391238] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-97 on buffer index:46
[ 323.393713] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-98 on buffer index:47
[ 323.396274] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-99 on buffer index:48
[ 323.398730] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-100 on buffer index:4
[ 332.968574] ###[consumer_thread_function]###[kConsumer-1] Consumer Thread stopped.
[ 332.970936] ###[thread_exit_module]###Total number of items produced: 100
[ 332.970942] ###[thread_exit_module]###Total number of items consumed: 100
[ 332.972970] ###[thread_exit_module]###The total elapsed time UID 1000 is 0:9:45
[ 332.976427] ###[thread_exit_module]###CSE330 Project Kernel Module Removed
Total Elapsed Time from PS Command 0:9:19
~/cse330-files-project2 #

```

Ln 1, Col 1 Tab Size: 4 UTF-8 CRLF {} C Layout: US Linux

./test.sh 500 50 1 1 100:

```
~/cse330-files-project2 # ./test.sh 500 50 1 1 100
Checking passed arguments
useradd: user 'test_cse330' already exists
/dev/null exists; moving on..
rm process_generator
gcc -o process_generator process_generator.c
Test script will start 500 processes for the user 1000
UID:1000 Process-5 Pid 1333
UID:1000 Process-11 Pid 1339
UID:1000 Process-1 Pid 1329
UID:1000 Process-6 Pid 1334
UID:1000 Process-17 Pid 1345
UID:1000 Process-30 Pid 1358
UID:1000 Process-13 Pid 1341
UID:1000 Process-34 Pid 1362
UID:1000 Process-4 Pid 1332
UID:1000 Process-35 Pid 1363
UID:1000 Process-3 Pid 1331
UID:1000 Process-36 Pid 1364
UID:1000 Process-12 Pid 1340
UID:1000 Process-37 Pid 1365
UID:1000 Process-38 Pid 1366
UID:1000 Process-40 Pid 1368
UID:1000 Process-42 Pid 1370
UID:1000 Process-44 Pid 1372
UID:1000 Process-46 Pid 1374
UID:1000 Process-47 Pid 1375
UID:1000 Process-49 Pid 1377
UID:1000 Process-50 Pid 1378
UID:1000 Process-51 Pid 1379
UID:1000 Process-52 Pid 1380
UID:1000 Process-55 Pid 1383
UID:1000 Process-57 Pid 1385
UID:1000 Process-59 Pid 1387
```

```
UID:1000 Process-491 Pid 1819
UID:1000 Process-492 Pid 1820
UID:1000 Process-493 Pid 1821
UID:1000 Process-494 Pid 1822
UID:1000 Process-495 Pid 1823
UID:1000 Process-496 Pid 1824
UID:1000 Process-497 Pid 1825
UID:1000 Process-498 Pid 1826
UID:1000 Process-499 Pid 1827
Continuing ..
Test script will insert the test kernel module
[ 875.366353] ###[thread_init_module]###CSE330 Project Kernel Module Inserted
[ 875.368132] ###[thread_init_module]###Kernel module received these inputs: UID:1000, Buffer-Si1
[ 875.371167] ###[producer_thread_function]###[kProducer-1] Produce-Item#:1 at buffer index: 0 f
[ 875.373382] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#:1 on buffer index:0::
[ 875.376765] ###[producer_thread_function]###[kProducer-1] Produce-Item#:2 at buffer index: 1 f
[ 875.376781] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#:2 on buffer index:1::
[ 875.379177] ###[producer_thread_function]###[kProducer-1] Produce-Item#:3 at buffer index: 2 f
[ 875.382638] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#:3 on buffer index:2::
[ 875.385551] ###[producer_thread_function]###[kProducer-1] Produce-Item#:4 at buffer index: 3 f
[ 875.389075] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#:4 on buffer index:3::
[ 875.393367] ###[producer_thread_function]###[kProducer-1] Produce-Item#:5 at buffer index: 4 f
The test script [ 875.395407] ###[producer_thread_function]###[kProducer-1] Produce-Item#:6 at b
waits 10 seconds[ 875.397816] ###[producer_thread_function]###[kProducer-1] Produce-Item#:7 at b
for the kernel [ 875.400398] ###[producer_thread_function]###[kProducer-1] Produce-Item#:8 at b
module to finish[ 875.403018] ###[producer_thread_function]###[kProducer-1] Produce-Item#:9 at b
[ 875.403543] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#:5 on buffer index:4::
[ 875.408930] ###[producer_thread_function]###[kProducer-1] Produce-Item#:10 at buffer index: 9
[ 875.408936] ###[producer_thread_function]###[kProducer-1] Produce-Item#:11 at buffer index: 10
[ 875.408938] ###[producer_thread_function]###[kProducer-1] Produce-Item#:12 at buffer index: 11
[ 875.411891] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#:6 on buffer index:5::
[ 875.416749] ###[producer_thread_function]###[kProducer-1] Produce-Item#:13 at buffer index: 12
[ 875.416754] ###[producer_thread_function]###[kProducer-1] Produce-Item#:14 at buffer index: 13
[ 875.416756] ###[producer_thread_function]###[kProducer-1] Produce-Item#:15 at buffer index: 14
[ 875.416758] ###[producer_thread_function]###[kProducer-1] Produce-Item#:16 at buffer index: 15

[ 875.416760] ###[producer_thread_function]###[kProducer-1] Produce-Item#:17 at buffer index: 16
```



```
[ 878.007490] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-487 on buffer index:3
[ 878.010537] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-488 on buffer index:3
[ 878.013620] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-489 on buffer index:3
[ 878.016632] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-490 on buffer index:3
[ 878.019625] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-491 on buffer index:4
[ 878.022674] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-492 on buffer index:4
[ 878.025838] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-493 on buffer index:4
[ 878.028872] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-494 on buffer index:4
[ 878.031862] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-495 on buffer index:4
[ 878.034879] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-496 on buffer index:4
[ 878.037884] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-497 on buffer index:4
[ 878.040923] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-498 on buffer index:4
[ 878.043929] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-499 on buffer index:4
[ 878.047039] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-500 on buffer index:4
```

Test script will remove the test kernel module

```
[ 885.440527] ###[consumer_thread_function]###[kConsumer-1] Consumer Thread stopped.
[ 885.442325] ###[thread_exit_module]###Total number of items produced: 500
[ 885.442330] ###[thread_exit_module]###Total number of items consumed: 500
[ 885.444016] ###[thread_exit_module]###The total elapsed time UID 1000 is      1:17:47
[ 885.447481] ###[thread_exit_module]###CSE330 Project Kernel Module Removed
```

The following is the dmesg output

```
[ 877.550234] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-438 on buffer index:3
[ 877.553016] ###[producer_thread_function]###[kProducer-1] Produce-Item#:470 at buffer index: 1
[ 877.555856] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-439 on buffer index:3
[ 877.558799] ###[producer_thread_function]###[kProducer-1] Produce-Item#:471 at buffer index: 2
[ 877.561958] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-440 on buffer index:3
[ 877.564599] ###[producer_thread_function]###[kProducer-1] Produce-Item#:472 at buffer index: 2
[ 877.564604] ###[producer_thread_function]###[kProducer-1] Produce-Item#:473 at buffer index: 2
[ 877.564606] ###[producer_thread_function]###[kProducer-1] Produce-Item#:474 at buffer index: 2
[ 877.564608] ###[producer_thread_function]###[kProducer-1] Produce-Item#:475 at buffer index: 2
[ 877.564610] ###[producer_thread_function]###[kProducer-1] Produce-Item#:476 at buffer index: 2
[ 877.564612] ###[producer_thread_function]###[kProducer-1] Produce-Item#:477 at buffer index: 2
[ 877.564614] ###[producer_thread_function]###[kProducer-1] Produce-Item#:478 at buffer index: 2
[ 877.564616] ###[producer_thread_function]###[kProducer-1] Produce-Item#:479 at buffer index: 2
[ 877.567296] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-441 on buffer index:4
[ 877.570007] ###[producer_thread_function]###[kProducer-1] Produce-Item#:480 at buffer index: 2
[ 877.572442] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-442 on buffer index:4
```



```
[ 877.968098] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-474 on buffer index:2
[ 877.971189] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-475 on buffer index:2
[ 877.974225] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-476 on buffer index:2
[ 877.977258] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-477 on buffer index:2
[ 877.980261] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-478 on buffer index:2
[ 877.983245] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-479 on buffer index:2
[ 877.986221] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-480 on buffer index:2
[ 877.989204] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-481 on buffer index:3
[ 877.992191] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-482 on buffer index:3
[ 877.995183] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-483 on buffer index:3
[ 877.998389] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-484 on buffer index:3
[ 878.001411] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-485 on buffer index:3
[ 878.004463] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-486 on buffer index:3
[ 878.007490] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-487 on buffer index:3
[ 878.010537] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-488 on buffer index:3
[ 878.013620] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-489 on buffer index:3
[ 878.016632] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-490 on buffer index:3
[ 878.019625] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-491 on buffer index:4
[ 878.022674] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-492 on buffer index:4
[ 878.025838] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-493 on buffer index:4
[ 878.028872] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-494 on buffer index:4
[ 878.031862] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-495 on buffer index:4
[ 878.034879] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-496 on buffer index:4
[ 878.037884] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-497 on buffer index:4
[ 878.040923] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-498 on buffer index:4
[ 878.043929] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-499 on buffer index:4
[ 878.047039] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-500 on buffer index:4
[ 885.440527] ###[consumer_thread_function]###[kConsumer-1] Consumer Thread stopped.
[ 885.442325] ###[thread_exit_module]###Total number of items produced: 500
[ 885.442330] ###[thread_exit_module]###Total number of items consumed: 500
[ 885.444016] ###[thread_exit_module]###The total elapsed time UID 1000 is      1:17:47
[ 885.447481] ###[thread_exit_module]###CSE330 Project Kernel Module Removed
Total Elapsed Time from PS Command 1:1:56
~/cse330-files-project2 #
```

[./test.sh](#) 100 50 1 2 25:

```
~/cse330-files-project2 # ./test.sh 100 50 1 2 25
Checking passed arguments
useradd: user 'test_cse330' already exists
/dev/null exists; moving on..
rm process_generator
gcc -o process_generator process_generator.c
Test script will start 100 processes for the user 1000
UID:1000 Process-4 Pid 1203
UID:1000 Process-10 Pid 1209
UID:1000 Process-3 Pid 1202
UID:1000 Process-1 Pid 1200
UID:1000 Process-2 Pid 1201
UID:1000 Process-5 Pid 1204
UID:1000 Process-7 Pid 1206
UID:1000 Process-9 Pid 1208
UID:1000 Process-8 Pid 1207
UID:1000 Process-6 Pid 1205
UID:1000 Process-11 Pid 1210
UID:1000 Process-12 Pid 1211
UID:1000 Process-13 Pid 1212
UID:1000 Process-14 Pid 1213
UID:1000 Process-15 Pid 1214
UID:1000 Process-16 Pid 1215
UID:1000 Process-17 Pid 1216
UID:1000 Process-18 Pid 1217
UID:1000 Process-19 Pid 1218
UID:1000 Process-20 Pid 1219
UID:1000 Process-21 Pid 1220
UID:1000 Process-22 Pid 1221
UID:1000 Process-23 Pid 1222
UID:1000 Process-24 Pid 1223
```

Ln 1, Col 1 Tab Size: 4 UTF-8 CRLF {} C Layout: US Linux

```

UID:1000 Process-88 Pid 1287
UID:1000 Process-89 Pid 1288
UID:1000 Process-90 Pid 1289
UID:1000 Process-91 Pid 1290
UID:1000 Process-92 Pid 1291
UID:1000 Process-93 Pid 1292
UID:1000 Process-94 Pid 1293
UID:1000 Process-95 Pid 1294
UID:1000 Process-96 Pid 1295
UID:1000 Process-97 Pid 1296
UID:1000 Process-98 Pid 1297
UID:1000 Process-99 Pid 1298
Continuing ..
Test script will insert the test kernel module
[ 544.891963] ###[thread_init_module]###CSE330 Project Kernel Module Inserted
[ 544.893553] ###[thread_init_module]###Kernel module received these inputs: UID:1000, Buffer-Si2
[ 544.896229] ###[producer_thread_function]###[kProducer-1] Produce-Item#:1 at buffer index: 0 f
[ 544.898711] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-1 on buffer index:0::
[ 544.898971] ###[producer_thread_function]###[kProducer-1] Produce-Item#:2 at buffer index: 1 f
[ 544.901965] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-2 on buffer index:1::
[ 544.904706] ###[producer_thread_function]###[kProducer-1] Produce-Item#:3 at buffer index: 2 f
[ 544.910204] ###[consumer_thread_function]###[kConsumer-2] Consumed Item#-1 on buffer index:2::
[ 544.910702] ###[producer_thread_function]###[kProducer-1] Produce-Item#:4 at buffer index: 3 f
[ 544.912758] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-3 on buffer index:3::
[ 544.914794] ###[producer_thread_function]###[kProducer-1] Produce-Item#:5 at buffer index: 4 f
[ 544.919410] ###[producer_thread_function]###[kProducer-1] Produce-Item#:6 at buffer index: 5 f
[ 544.919471] ###[consumer_thread_function]###[kConsumer-2] Consumed Item#-2 on buffer index:4::
[ 544.925179] ###[producer_thread_function]###[kProducer-1] Produce-Item#:7 at buffer index: 6 f
The test script [ 544.927279] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-4 on
waits 10 seconds for the kernel module to finish[ 544.932351] ###[consumer_thread_function]###[k
[ 544.936780] ###[producer_thread_function]###[kProducer-1] Produce-Item#:8 at buffer index: 7 f
[ 544.937709] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-5 on buffer index:7::
[ 544.940092] ###[producer_thread_function]###[kProducer-1] Produce-Item#:9 at buffer index: 8 f
[ 544.942933] ###[consumer_thread_function]###[kConsumer-2] Consumed Item#-4 on buffer index:8::
[ 544.946454] ###[producer_thread_function]###[kProducer-1] Produce-Item#:10 at buffer index: 9
[ 544.948792] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-6 on buffer index:9::

```

```

[ 545.362162] ####[consumer_thread_function]####[kConsumer-2] Consumed Item#-54 on buffer index:43
[ 545.364339] ####[consumer_thread_function]####[kConsumer-1] Consumed Item#-41 on buffer index:44
[ 545.366636] ####[consumer_thread_function]####[kConsumer-2] Consumed Item#-55 on buffer index:45
[ 545.368748] ####[consumer_thread_function]####[kConsumer-1] Consumed Item#-42 on buffer index:46
[ 545.368752] ####[consumer_thread_function]####[kConsumer-1] Consumed Item#-43 on buffer index:47
[ 545.368754] ####[consumer_thread_function]####[kConsumer-1] Consumed Item#-44 on buffer index:48
[ 545.368756] ####[consumer_thread_function]####[kConsumer-1] Consumed Item#-45 on buffer index:49
Test script will remove the test kernel module
[ 554.934031] ####[consumer_thread_function]####[kConsumer-1] Consumer Thread stopped.
[ 554.935764] ####[consumer_thread_function]####[kConsumer-2] Consumer Thread stopped.
[ 554.937523] ####[thread_exit_module]####Total number of items produced: 100
[ 554.937527] ####[thread_exit_module]####Total number of items consumed: 100
[ 554.938968] ####[thread_exit_module]####The total elapsed time UID 1000 is      0:9:47
[ 554.942171] ####[thread_exit_module]####CSE330 Project Kernel Module Removed
The following is the dmesg output
[ 545.344607] ####[producer_thread_function]####[kProducer-1] Produce-Item#:95 at buffer index: 44
[ 545.344611] ####[producer_thread_function]####[kProducer-1] Produce-Item#:96 at buffer index: 45
[ 545.344614] ####[producer_thread_function]####[kProducer-1] Produce-Item#:97 at buffer index: 46
[ 545.347540] ####[consumer_thread_function]####[kConsumer-2] Consumed Item#-51 on buffer index:38
[ 545.349817] ####[producer_thread_function]####[kProducer-1] Produce-Item#:98 at buffer index: 47
[ 545.352688] ####[consumer_thread_function]####[kConsumer-1] Consumed Item#-39 on buffer index:39
[ 545.354857] ####[producer_thread_function]####[kProducer-1] Produce-Item#:99 at buffer index: 48
[ 545.357389] ####[consumer_thread_function]####[kConsumer-2] Consumed Item#-52 on buffer index:40
[ 545.359496] ####[producer_thread_function]####[kProducer-1] Produce-Item#:100 at buffer index: 4
[ 545.359501] ####[producer_thread_function]####[kProducer-1] Producer Thread stopped.
[ 545.359546] ####[consumer_thread_function]####[kConsumer-1] Consumed Item#-40 on buffer index:41
[ 545.362157] ####[consumer_thread_function]####[kConsumer-2] Consumed Item#-53 on buffer index:42
[ 545.362162] ####[consumer_thread_function]####[kConsumer-2] Consumed Item#-54 on buffer index:43
[ 545.364339] ####[consumer_thread_function]####[kConsumer-1] Consumed Item#-41 on buffer index:44
[ 545.366636] ####[consumer_thread_function]####[kConsumer-2] Consumed Item#-55 on buffer index:45
[ 545.368748] ####[consumer_thread_function]####[kConsumer-1] Consumed Item#-42 on buffer index:46
[ 545.368752] ####[consumer_thread_function]####[kConsumer-1] Consumed Item#-43 on buffer index:47
[ 545.368754] ####[consumer_thread_function]####[kConsumer-1] Consumed Item#-44 on buffer index:48
[ 545.368756] ####[consumer_thread_function]####[kConsumer-1] Consumed Item#-45 on buffer index:49
[ 554.934031] ####[consumer_thread_function]####[kConsumer-1] Consumer Thread stopped.
[ 554.935764] ####[consumer_thread_function]####[kConsumer-2] Consumer Thread stopped.
[ 554.937523] ####[thread_exit_module]####Total number of items produced: 100

```

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[ 545.368754] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-44 on buffer index:48
[ 545.368756] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-45 on buffer index:49
Test script will remove the test kernel module
[ 554.934031] ###[consumer_thread_function]###[kConsumer-1] Consumer Thread stopped.
[ 554.935764] ###[consumer_thread_function]###[kConsumer-2] Consumer Thread stopped.
[ 554.937523] ###[thread_exit_module]###Total number of items produced: 100
[ 554.937527] ###[thread_exit_module]###Total number of items consumed: 100
[ 554.938968] ###[thread_exit_module]###The total elapsed time UID 1000 is 0:9:47
[ 554.942171] ###[thread_exit_module]###CSE330 Project Kernel Module Removed
The following is the dmesg output
[ 545.344607] ###[producer_thread_function]###[kProducer-1] Produce-Item#:95 at buffer index: 44
[ 545.344611] ###[producer_thread_function]###[kProducer-1] Produce-Item#:96 at buffer index: 45
[ 545.344614] ###[producer_thread_function]###[kProducer-1] Produce-Item#:97 at buffer index: 46
[ 545.347540] ###[consumer_thread_function]###[kConsumer-2] Consumed Item#-51 on buffer index:38
[ 545.349817] ###[producer_thread_function]###[kProducer-1] Produce-Item#:98 at buffer index: 47
[ 545.352688] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-39 on buffer index:39
[ 545.354857] ###[producer_thread_function]###[kProducer-1] Produce-Item#:99 at buffer index: 48
[ 545.357389] ###[consumer_thread_function]###[kConsumer-2] Consumed Item#-52 on buffer index:40
[ 545.359496] ###[producer_thread_function]###[kProducer-1] Produce-Item#:100 at buffer index: 4
[ 545.359501] ###[producer_thread_function]###[kProducer-1] Producer Thread stopped.
[ 545.359546] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-40 on buffer index:41
[ 545.362157] ###[consumer_thread_function]###[kConsumer-2] Consumed Item#-53 on buffer index:42
[ 545.362162] ###[consumer_thread_function]###[kConsumer-2] Consumed Item#-54 on buffer index:43
[ 545.364339] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-41 on buffer index:44
[ 545.366636] ###[consumer_thread_function]###[kConsumer-2] Consumed Item#-55 on buffer index:45
[ 545.368748] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-42 on buffer index:46
[ 545.368752] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-43 on buffer index:47
[ 545.368754] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-44 on buffer index:48
[ 545.368756] ###[consumer_thread_function]###[kConsumer-1] Consumed Item#-45 on buffer index:49
[ 554.934031] ###[consumer_thread_function]###[kConsumer-1] Consumer Thread stopped.
[ 554.935764] ###[consumer_thread_function]###[kConsumer-2] Consumer Thread stopped.
[ 554.937523] ###[thread_exit_module]###Total number of items produced: 100
[ 554.937527] ###[thread_exit_module]###Total number of items consumed: 100
[ 554.938968] ###[thread_exit_module]###The total elapsed time UID 1000 is 0:9:47
[ 554.942171] ###[thread_exit_module]###CSE330 Project Kernel Module Removed
Total Elapsed Time from PS Command 0:9:12
~/cse330-files-project2 #

```